

Genetics — Complete Notes

Grade 12

Life Sciences

CAPS

What you'll learn

- the meaning of the key genetics terms, such as gene, allele, genotype and phenotype
- Mendel's three laws of inheritance
- how to set out and read a monohybrid cross
- how to set out and read a dihybrid cross
- incomplete dominance, co-dominance and multiple alleles
- how sex is determined and how sex-linked traits are inherited
- how to use a test cross and read a pedigree
- how to answer genetics problems step by step

Genetics — the basics

Genetics is the study of heredity: how characteristics (traits) are passed from parents to offspring. The instructions for these traits are carried by genes.

Term	Meaning
Gene	A section of DNA that codes for one protein, and therefore for a particular characteristic
Allele	One of the different forms of a gene (for example, T = tall, t = short)
Chromosome	A thread-like structure of DNA and protein that carries many genes
Genotype	The genetic make-up, that is the alleles present, e.g. Tt
Phenotype	The physical appearance of an organism, e.g. tall
Homozygous	Having two identical alleles for a gene, e.g. TT or tt
Heterozygous	Having two different alleles for a gene, e.g. Tt
Dominant allele	An allele that is expressed even when only one copy is present (written with a capital letter)
Recessive allele	An allele that is expressed only when two copies are present (written with a small letter)
Pure-breeding	An organism that is homozygous for a trait and always passes on the same allele
Gamete	A sex cell (sperm, egg or pollen) that carries one allele of each gene
Carrier	A heterozygous organism that has a recessive allele but does not show the recessive phenotype

Mendel's laws

Gregor Mendel, an Austrian monk who worked with pea plants, is called the father of genetics. From his experiments he formulated three laws:

1. Law of Dominance: when two different alleles for a gene are present, the dominant allele masks the recessive allele, so only the dominant phenotype is seen.
2. Law of Segregation: each organism has two alleles for each gene, and during the formation of gametes these alleles separate (segregate), so each gamete receives only one allele.
3. Law of Independent Assortment: during gamete formation the alleles of different genes are distributed to the gametes independently of one another.

Monohybrid crosses

A monohybrid cross follows one characteristic controlled by one gene with two alleles.

Worked example 1 — two heterozygous parents (Tt x Tt)

Problem: In pea plants, tall (T) is dominant over short (t). Two heterozygous tall plants are crossed. Determine the possible genotypes and phenotypes of the offspring.

Step 1 — write down the genotypes of the parents. Parent 1 is Tt (tall) and Parent 2 is Tt (tall).

Step 2 — work out the gametes. Each gamete gets ONE allele, so a Tt parent makes T and t gametes.

Step 3 — draw a Punnett square. Put the alleles of one parent across the top and of the other parent down the side, then combine them in each block.

	T	t	
	+-----+	+-----+	
T	TT	Tt	
	+-----+	+-----+	
t	Tt	tt	
	+-----+	+-----+	

Step 4 — read the results. The genotypes are 1 TT : 2 Tt : 1 tt, so the genotype ratio is 1 : 2 : 1. The phenotypes are 3 tall (TT and Tt) : 1 short (tt), so the phenotype ratio is 3 : 1.

Step 5 — answer in words. On average, three out of every four offspring are tall and one out of four is short, so there is a 75% chance of a tall offspring and a 25% chance of a short offspring.

Worked example 2 — a test cross

Problem: A tall pea plant could be TT or Tt; you cannot tell by looking at it. Explain how a test cross can reveal its genotype.

Step 1 — cross the unknown plant with a homozygous recessive plant (tt, short).

Step 2 — predict the two possible outcomes.

If the unknown plant is TT:

		T		T	
		+-----+	+-----+		
t		Tt		Tt	
		+-----+	+-----+		
t		Tt		Tt	
		+-----+	+-----+		

All offspring are Tt, so 100% are tall.

If the unknown plant is Tt:

		T		t	
		+-----+	+-----+		
t		Tt		tt	
		+-----+	+-----+		
t		Tt		tt	
		+-----+	+-----+		

The offspring are 50% tall (Tt) and 50% short (tt).

Step 3 — interpret the result. If any short offspring appear, the unknown parent must be heterozygous (Tt). If all the offspring are tall, the parent is most likely homozygous (TT), although a very small number of offspring could hide a heterozygous parent.

Dihybrid crosses

A dihybrid cross follows two characteristics at the same time, controlled by two different genes.

Worked example 3 — RrYy x RrYy (a full dihybrid cross)

Problem: In pea plants, round seed (R) is dominant over wrinkled (r), and yellow seed (Y) is dominant over green (y). Two plants that are heterozygous for both traits (RrYy) are crossed. Determine the phenotype ratio of the offspring.

Step 1 — the parents are RrYy and RrYy.

Step 2 — work out the gametes. Each gamete gets one allele of each gene, so an RrYy plant produces four kinds of gamete: RY, Ry, rY and ry.

Step 3 — draw a 4 x 4 Punnett square with 16 blocks.

	RY	Ry	rY	ry
	+-----+	+-----+	+-----+	+-----+
RY	RRYY	RRYy	RrYY	RrYy
	+-----+	+-----+	+-----+	+-----+
Ry	RRYy	RRyy	RrYy	Rryy
	+-----+	+-----+	+-----+	+-----+
rY	RrYY	RrYy	rrYY	rrYy
	+-----+	+-----+	+-----+	+-----+
ry	RrYy	Rryy	rrYy	rryy
	+-----+	+-----+	+-----+	+-----+

Step 4 — group the sixteen offspring by phenotype.

- Round and yellow (at least one R and one Y): RRYY, RRYy, RrYY, RrYy, RRYy, RrYY, RrYy, RrYy = 9 offspring
- Round and green (at least one R, and yy): RRyy, Rryy, Rryy = 3 offspring
- Wrinkled and yellow (rr, and at least one Y): rrYY, rrYy, rrYy = 3 offspring
- Wrinkled and green (rryy): rryy = 1 offspring

Step 5 — answer. The phenotype ratio is 9 : 3 : 3 : 1 (round yellow : round green : wrinkled yellow : wrinkled green).

Incomplete dominance and co-dominance

Definition — incomplete dominance: neither allele is completely dominant, so the heterozygous phenotype is a blend (an intermediate) of the two homozygous phenotypes.

Example: a red snapdragon (RR) crossed with a white snapdragon (WW) gives all pink offspring (RW). When two pink plants are crossed (RW x RW):

	R	W
	+-----+	+-----+
R	RR	RW
	+-----+	+-----+
W	RW	WW
	+-----+	+-----+

The result is 1 red (RR) : 2 pink (RW) : 1 white (WW), so the phenotype ratio 1 : 2 : 1 is the same as the genotype ratio.

Definition — co-dominance: both alleles are expressed fully and equally in the heterozygote, so both phenotypes appear side by side. Examples include roan cattle (RW, with both red and white hairs) and human blood group AB, where both the A and B antigens are present.

Multiple alleles — the ABO blood groups

The ABO blood group is controlled by one gene that has three alleles: I^A , I^B and i . Here I^A and I^B are co-dominant (both are expressed when present together), and both are dominant over i .

Genotype	Blood group
IA IA or IA i	A
IB IB or IB i	B
IA IB	AB
ii	O

A person with blood group O has no A or B antigens on the red blood cells, so O blood can be given to anyone — O is called the universal donor. A person with blood group AB has no anti-A or anti-B antibodies in the plasma, so AB blood can receive blood from any group — AB is called the universal recipient.

Sex determination and sex-linked traits

Humans have 23 pairs of chromosomes: 22 pairs of autosomes and 1 pair of sex chromosomes. Females are XX and males are XY.

- The mother can pass on only an X chromosome (all her eggs carry X).
- The father passes on either an X or a Y (his sperm are about 50% X and 50% Y).
- If a sperm carrying X fertilises the egg, the child is XX (female). If a sperm carrying Y fertilises the egg, the child is XY (male). The father therefore determines the sex of the child.

Sex-linked traits are controlled by genes on the sex chromosomes, mostly on the X chromosome. Because a male has only one X chromosome, a single recessive allele on that X is enough to show the condition, so males are affected more often than females. Examples include red-green colour blindness and haemophilia.

In these notes, XH is the normal allele and Xh is the recessive allele for the condition (both are carried on the X chromosome). The Y chromosome carries no matching allele.

Worked example 4 — a carrier mother and an unaffected father

Problem: A woman is a carrier of haemophilia (XH Xh) and her partner is unaffected (XH Y). Determine the possible genotypes and phenotypes of their children.

Step 1 — write the parents' genotypes: mother XH Xh, father XH Y.

Step 2 — work out the gametes. The mother gives XH or Xh; the father gives XH or Y.

Step 3 — draw a Punnett square.

	XH	Xh	
	+-----+	+-----+	
XH	XHXH	XHXh	
	+-----+	+-----+	
Y	XHY	XhY	
	+-----+	+-----+	

Step 4 — read the results.

- XHXH: normal daughter, not a carrier (25% of all children)
- XHXh: carrier daughter (25% of all children)
- XHY: unaffected son (25% of all children)

- XhY: affected son with haemophilia (25% of all children)

Step 5 — answer. All daughters are unaffected, but half of them are carriers. Half of the sons are unaffected and half have haemophilia. Overall, there is a 25% chance that any child is an affected son.

Pedigrees and test crosses

Definition — pedigree: a family-tree diagram that shows how a trait is inherited over several generations. Squares represent males and circles represent females; a shaded symbol is an affected individual.

- If a trait appears in every generation and affects both sexes, it may be dominant.
- If a trait skips generations, it is usually recessive.
- If mostly males are affected, the trait may be sex-linked.
- If two unaffected parents have an affected child, the trait is recessive and both parents are carriers.

A test cross is used to find an unknown genotype. The unknown individual is crossed with a homozygous recessive individual, and the phenotypes of the offspring reveal the unknown genotype.

Common mistakes

- Confusing genotype with phenotype: the genotype is the alleles (e.g. Tt) and the phenotype is the appearance (e.g. tall).
- Using different letters for the two alleles of one gene. Use the same letter, with a capital for the dominant allele and a small letter for the recessive allele (Tt, not Td).
- Forgetting that each gamete carries only ONE allele of each gene, so a Tt plant produces T and t gametes, not Tt.
- Mixing up IA, IB and i in blood-group problems.
- Saying that a carrier has the disorder. A carrier has the recessive allele but does not show the recessive phenotype.
- Forgetting that the father determines the sex of the baby because he provides either an X or a Y.
- Treating a 3 : 1 ratio as an exact result. It is a probability, so real families can differ.
- In a dihybrid cross, listing only two gametes instead of four (RY, Ry, rY, ry).
- Confusing incomplete dominance (a blended phenotype) with co-dominance (both phenotypes shown).

Quick summary / cheat sheet

- A gene is a section of DNA for a protein; an allele is one form of a gene.
- A dominant allele masks a recessive allele, so only the capital-letter phenotype shows in a heterozygote.
- Monohybrid Tt x Tt gives a genotype ratio of 1 : 2 : 1 and a phenotype ratio of 3 : 1.
- Dihybrid RrYy x RrYy gives a phenotype ratio of 9 : 3 : 3 : 1.
- Incomplete dominance gives a blend (red x white gives pink).
- Co-dominance shows both phenotypes (blood group AB, roan cattle).
- The ABO blood group uses three alleles: IA, IB (co-dominant) and i.
- The father determines sex: XX is female and XY is male.
- X-linked recessive traits such as haemophilia affect males more often because males have only one X.
- A test cross uses a homozygous recessive partner to reveal an unknown genotype.

Practice questions

1. Define the terms allele and phenotype.
2. In guinea pigs, black fur (B) is dominant over white fur (b). Two black guinea pigs (Bb) are crossed. Use a Punnett square to give the genotype and phenotype ratios of the offspring.
3. A plant with genotype RrYy is self-pollinated. List the four types of gamete it can produce.
4. Explain why a person with blood group O is called a universal donor.
5. A colour-blind man marries a woman who is homozygous normal (X^H X^H). What is the chance that their sons are colour blind?
6. Give the phenotype ratio expected from the cross RW x RW, where R and W show incomplete dominance.
7. State TWO differences between co-dominance and incomplete dominance.
8. A tall pea plant of unknown genotype is crossed with a short (tt) plant, and half of the offspring are short. Give the genotype of the tall plant and explain your answer.

Answers

1. An allele is one of the different forms of a gene (for example, T or t for height). A phenotype is the observable physical characteristic of an organism (for example, tall).
2. The parents are Bb x Bb and their gametes are B and b.

	B		b	
	+-----+		+-----+	
B		BB		Bb
	+-----+		+-----+	
b		Bb		bb
	+-----+		+-----+	

The genotype ratio is 1 BB : 2 Bb : 1 bb and the phenotype ratio is 3 black : 1 white.

3. The four gametes are RY, Ry, rY and ry.
4. A person with blood group O has neither A nor B antigens on the red blood cells, so the recipient's immune system has nothing to react against — this means O blood can safely be given to people of any blood group.
5. The chance is 0%. A colour-blind man is X^h Y and the woman is X^H X^H, so every son receives X^H from his mother and is X^H Y (unaffected); the sons cannot inherit the X-linked allele from their father because he passes on his Y to sons.
6. The ratio is 1 red (RR) : 2 pink (RW) : 1 white (WW).
7. In incomplete dominance the heterozygote shows a blend of the two phenotypes (red x white gives pink), while in co-dominance both phenotypes are expressed fully and separately (blood group AB shows both A and B antigens). In incomplete dominance the phenotype ratio equals the genotype ratio, which is not the case for co-dominance.
8. The tall plant is heterozygous (Tt). A test cross with tt gives a Tt parent 50% Tt (tall) and 50% tt (short) offspring, which matches the recorded result; a TT parent would give only tall offspring.

Official EduRank Team notes. Always cross-check with your textbook and teacher.